Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/02/2004
PI Name:	Lelkes, Peter Istvan Ph.D.		
Project Title:	Differential Gene Expression in Endothelial Cells Cultured in Rotating Wall Vessel Bioreactors & Exposed to Low Laminar Shear Stress		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	19122-6018	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-08-B
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	Rotating Wall Vessel Bioreactors offer a unique biotechnology approach to 3-D cell culture and advanced tissue engineering applications. In these fluid-filled Bioreactors, particles suspended in the culture medium rotate with the fluid in a quasi-solid body configuration. Cells and cell assemblies in this venue are in continual free fall and thus concomitantly exposed to low laminar shear stress (~ 0.5 dynes/cm2) and also to "simulated or modeled microgravity ", i.e. to a time-averaged, randomized gravitational force (~0.01 x g). To date, the contribution of each of these factors, especially of reduced laminar shear stress, has not been thoroughly assessed experimentally. Based on reports in the literature and our own preliminary data, we hypothesize that the component which simulates/models microgravity by "randomizing" the gravitational vector, rather than low, laminar flow–induced shear stress, may be instrumental in modulating gene expression and functional differentiation of cells/cell assemblies in RWV Bioreactors. In testing our hypothesis, we propose to use genomics, proteomics and bioinformatics approaches for analyzing patterns of differential		

Task Description:	expression of genes and gene products in human endothelial cells cultured for various amounts of time (4 h –20 days) in RWV Bioreactors, with controls maintained static monolayers, static suspension cultures, or as monolayers in a parallel plate flow chamber exposed to fluid shear stress of 0.5 dynes/cm2. In addition, by using the Perfused Rotating Wall Vessel Reactors, we will be able induce controlled levels of fluid shear stress levels in these Bioreactors. Endothelial cells are chosen for this study, since a) the fluid-shear dependent gene activation in these cells is well established and characterized b) altered fluid dynamics and mechanical forces, such as present in RWV Bioreactors (and space flight?) are known risk factors in the etiology of diminished cardiac and cardiovascular function, specifically in the etiology of atherosclerosis. The objectives of this study are as follows: Specific aim 1: Compare differential gene expression in human microvascular endothelial cells cultured respectively, as monolayers under static conditions, or exposed to low levels of steady shear stress, and in suspension culture in Rotating Wall Vessel Bioreactors. Specific Aim 2: Compare differential protein expression in human microvascular endothelial cells cultured under the same conditions described above in Specific Aim1. Specific aim 3: Use modern bioinformatics approaches for identifying functional correlations between expression patterns of genes and gene products, specifically of signaling molecules and pathways, which uniquely are expressed (only) in RWV Bioreactors Specific Aim 4: Use established molecular biological approaches for identifying a unique RWV –sensitive response element in the promoter region of one or more RWV-responsive genes. . While being responsive to several different stipulations of the soliciting NRA (01-OBPR-08), the objectives of this ground-based proposal is of relevance to NASA's mission, as outlined in the Critical Path Roadmap, specifically, Risk 6 Diminished Cardiac and Vascular Function. In this context our study adresses Enbling Questions 6a-6c: Does long-duration space flight lead to diminished cardiac function? What mechanisms are involved? Is the process reversible? • In studying the workings of NASA developed RWV Bioreactor Biotechnology we are focusing on a) elucidating the value of low shear culturing and b) studying differential endothelial gene expression in RWV Bioreactors. • This study will distinguish between the effects of low shear and that of other parameters in the workings of RWV Bioreactors, specifically those components that may represent a ground-based microgravity analog. Furthermore, the comprehensive approach will lead to the identification of candidate mechanism and genes/gene products that are uniquely responsive to the modeled microgravity component of the RWV environment. • The outcome of this study will facilitate rationally designing meaningful space flight experiments, aimed at deciphering the mechanisms by which cells and organisms sense and transduce microgravity. Thus, this study provides a strong link and a clear path to flight research on the International Space Station and beyond for long-duration space flight. • The proposed studies entail strong educational components, both for our students as well as for students from local/regional schools.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed study will enable the rational use of an important flight hardware, the RWV Bioreactor system, for ground based studies addressing important risks that figure prominetly in the Critical Road Map, such as such as cardiac and vascular functions, impaired immune responses, etc. In addition, the strong educational component of this study will be levaraged for outreach activities, to raise the level of awareness and enthusiasm for NASA-related projects amongst high school and college students.
Task Progress:	In the first year of this grant we have made significant progress in the first thre of our four specific aims. We have conducted the initial set of experiments aimed at comparing differential gene expression in human microvascular endothelial cells cultured respectively, as monolayers under static conditions, or exposed to low levels of steady shear stress, and in suspension culture in Rotating Wall Vessel Bioreactor (specific aim 1). At this time we are mining our data in order to identify functional correlations between expression patterns of the genes which are differentially expressed under the various experimental conditions. A careful analysis of our initial data has revealed that amongst the ~ 20,000 human genes studied, some 3000 are differentially expressed in one or more of the culture venues. Importantly we have also identified a significant number (several hundred) of genes which seem to be uniquely are expressed (only) in RWV Bioreactors, and which may be related to the modeled microgravity component in this venue. Currently we are using sophisticated, up-to date bioinformatics approaches and databases (such as DAVID, KEGG, GO and Ingenuity) for elucidating some of the signaling molecules and pathways, which characterize the biological responses in each of the venues studied (specific aim3). In terms of specific aim 2, we are developing our in house 2-D capabilities. Using the Bio-Rad multiplex proteomics tools, we have improved our capabilities to reproducibly run 2-D gels. In the context of this proposal which we can reliably resolve > 1000 proteins. Hence we can now begin to compare differential protein expression in human microvascular endothelial cells cultured under the same conditions described above. Current and future efforts in this are will focus on analyzing the differential expression phosphoproteins, as a commensurate of cellular signaling. As before, we will use complex bioinformatics tools to elucidate differential activation of signaling pathways in the various venues.
Bibliography Type:	Description: (Last Updated: 09/28/2023)
Articles in Peer-reviewed Journals	Brooks AR, Lelkes PI, Rubanyi GM. "Gene expression profiling of vascular endothelial cells exposed to fluid mechanical forces: relevance for focal susceptibility to atherosclerosis." Endothelium. 2004 Jan-Feb;11(1):45-57. Review. PubMed PMID: 15203878 , Jan-2004
Presentation	Lelkes, P.I. "An integrated genomics and proteomics approach towards understanding differential activation of endothelial cells by distinct hemodynamic force" FASEB Summer Research Conference "Molecular and Cellular Signaling in the Perinatal Cardiovascular System", Tucson, AZ , August 2004 Aug-2004
Presentation	Lelkes, P.I. "A Biophysical Analysis of Vascular Pathophysiology: A Hemodynamic Etiology of Predisposition to Vascular Injury in Atherosclerosis and Hypertension" 4th International Conference on Physiological Biophysics, November 2004, Shanghai, PR China Nov-2004